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MOOC
Machine Learning in Weather & Climate

TRANSCRIPT

What is machine learning and what types of machine learning are there?

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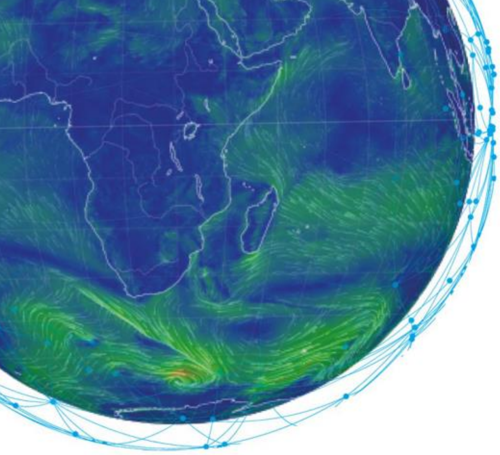

What is machine learning and what types of machine learning are there?

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Introduction

Hello and welcome. My name is Lisa Burke and today, we'll meet Dr. Jesper Dramsch, who will introduce the different types of machine learning and associated tasks with each type.

In this short video, Dr. Dramsch will explain what machine learning is and the various types of machine learning that exist.

Ready? So, let's go to ECMWF in Bonn in Germany.

Script

In modern days, machine learning and AI are often used interchangeably. However, machine learning has a very nice definition, within the wider field of AI. "Machine Learning is a set of algorithms that improve their performance on a set task through experience."

Often this is achieved by a combination of statistical methods and numerical optimization to incrementally improve the machine learning system and gain insight into the task that generalizes to future variations of that same task. This notion of a task is a little abstract, but it leads us right into the different types of machine learning.

The most commonly known type of machine learning is supervised learning. In supervised learning, we have a dataset and labels or output data for this dataset. Then we train a machine learning model to map the samples in our dataset to the according labels. Tasks within this type of machine learning are classification tasks, like detecting clouds in satellite images, or regression tasks, like predicting the temperature from observations at weather stations.

ML modelling is different to classical numerical models we build, where we have the data and the rules and obtain answers from the model. In supervised machine learning, we flip this notion on its head and provide data and answers to derive the rules (albeit those rules are often implicitly stated).

Another type of machine learning is defined in opposition to supervised learning: unsupervised learning. This type of learning is relevant for large datasets that do not come with labels or output data. Here machine learning can be used to explore the internal structure of the data to obtain insights.

Typical tasks in unsupervised machine learning are clustering data. This task consists of building groups of samples in the dataset that belong together based on similarities within the data itself. An example here would be grouping weather regimes globally to find similarities in regional patterns.

Another common task consists of reducing the complexity of data. Some datasets contain multiple complex measurements, which can be challenging to process as a human and even for computers. Atmospheric chemistry comes to mind, where many aerosols and molecules are recorded. This task, called dimensionality reduction, represents the original data, which is still as expressive but easier to visualize and parse for subsequent processes.

The next type of machine learning has recently gained more traction and sits between supervised and unsupervised machine learning: semi-supervised machine learning. This type relies on partially labelled datasets, and this is often the case for extremely large datasets, where assigning corresponding outputs for every sample would simply be too expensive. Semi-supervised algorithms then use both the internal structure of the data and the assigned outputs of some samples to transfer these labels to both the labelled and unlabelled portions of the data.

A possible task for this problem is cloud detection in satellite images. There we have an extensive amount of data, however, assigning cloud labels takes human interpreters a significant amount of time. Here, we can label clouds in a subset of the data and semi-supervised learning can still use the full dataset, despite a lack of complete labels.

The final type of machine learning is called reinforcement learning. This type is a little different to the other types we just learned about. Here we define rules and rewards in an environment for an algorithm to explore and learn from.

The most straightforward example task for this type of machine learning is game AI, like the recent advances in computers playing chess and Go. Other tasks include self-driving vehicles and autonomous robots. There are fewer example tasks in weather and climate, as these environments can be tricky to define for complex systems like the global atmospheric system. However, it has been used in decision systems for irrigation based on weather data.

These four types of machine learning give an overview of different opportunities for application in weather and climate prediction. Whenever a system can benefit from more data points to learn a better representation of our world, we can find a machine learning task that could model this connection.

Summary

In this video we learned about different applications for machine learning in weather and climate and saw examples how each can be used. We saw that supervised and unsupervised methods have already found strong applications within the field, whereas semi-supervised and reinforcement learning are still less explored.

Thank you, Jesper, for this great overview about the different types of machine learning. In the next video Peter Dueben will take us through the opportunities that arise from working with machine learning in weather and climate modelling. In the meantime, if you have any question or would like to discuss this subject in further detail, please use our forum on Moodle. It will be a pleasure to hear you!